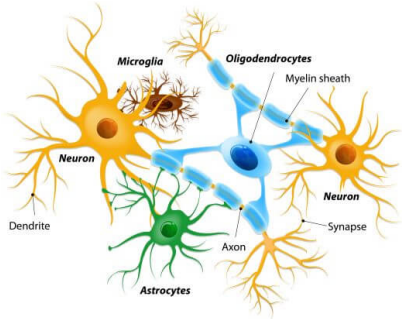
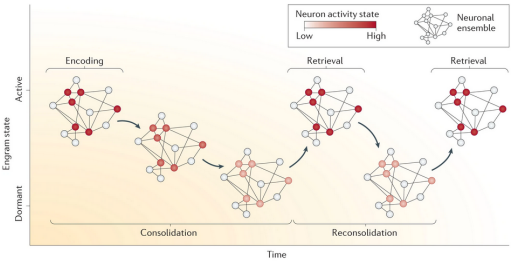
Biology | Paper

Memories may be physically stored in the hippograms as patterns of connections between neurons (engrams).

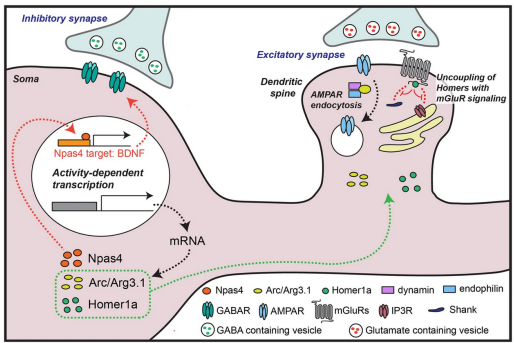






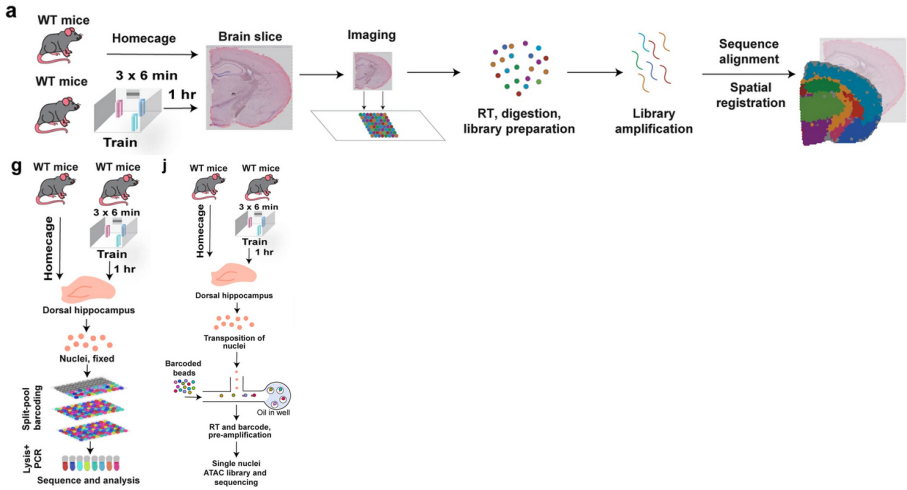
GENSAT, biology dictionary, Josselyn et al. (2015)

"Neuronal populations contributing to engram ensembles are activated [in part via IEGs] by learning and endure cellular changes …"



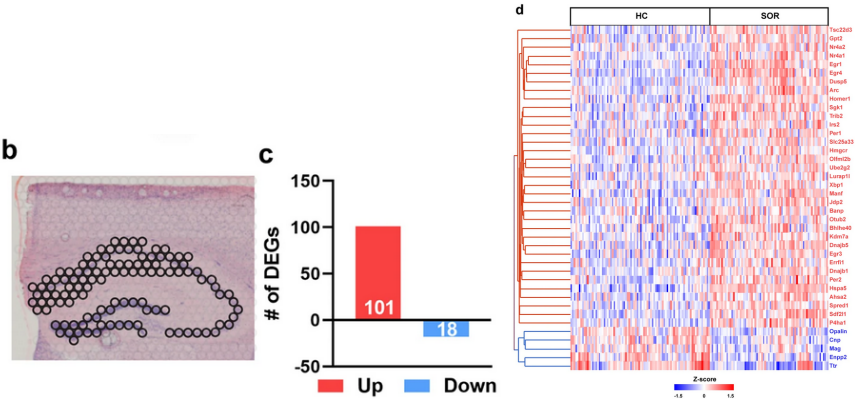
Kim et al. (2018)

- Question: What is the “activity-driven spatial transcriptomic diversity within the hippocampal network”?
- Why is this important: ?
- Why hasn’t this been done already: ?

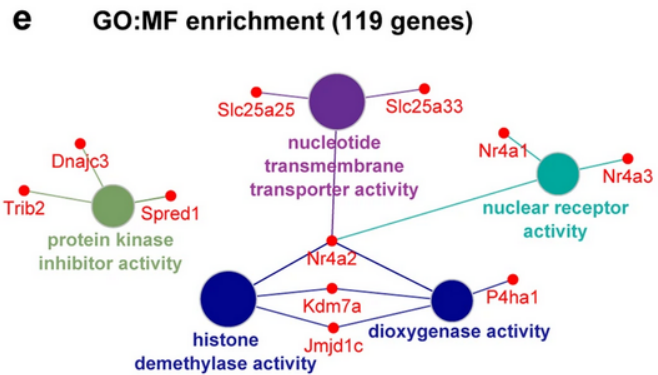

a
WT mice
Homecage
Brain slice
Imaging
WT mice
3 x 6 min
1 hr
Train
RT, digestion, library preparation
Library amplification
Sequence alignment
Spatial registration
g
WT mice
WT mice
3 x 6 min
Train
1 hr
Homecage
Dorsal hippocampus
Nuclei, fixed
Split-pool barcoding
Lysis+ PCR
Sequence and analysis
j
WT mice
WT mice
3 x 6 min
Train
1 hr
Homecage
Dorsal hippocampus
Transposition of nuclei
Barcoded beads
Oil in well
RT and barcode, pre-amplification
Single nuclei ATAC library and sequencing

Pseudobulk analysis of hippocampal spatial transcriptomics following learning correlates with bulk RNA sequencing

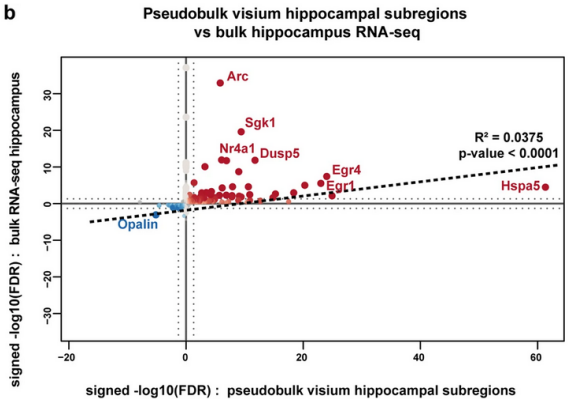
What genes become up- and down-regulated in as a result of spatial object recognition (SOR) training?



What biological processes are involved in SOR training?

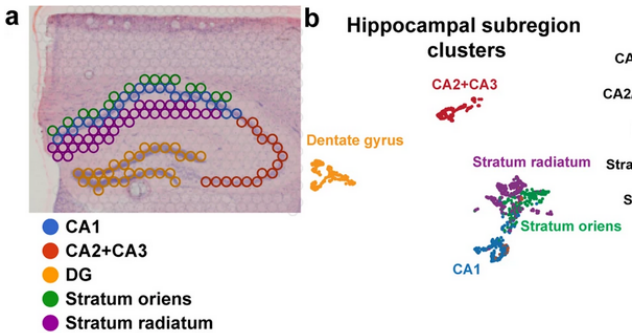


How consistent are the pseudobulk and bulk RNA-seq results?

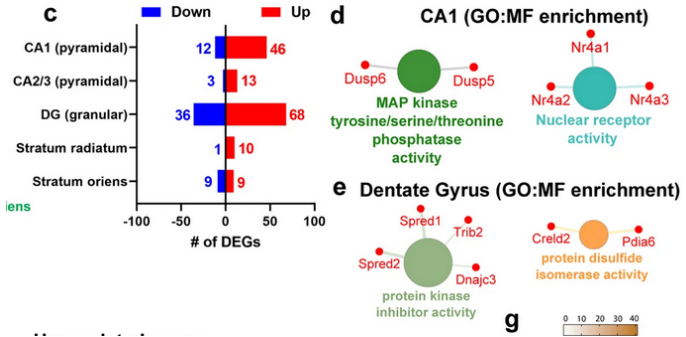


Hippocampal subregions exhibit distinct transcriptomic signatures following learning

To what extent are spatially distinct subregions of the hippocampus also transcriptionally distinct?

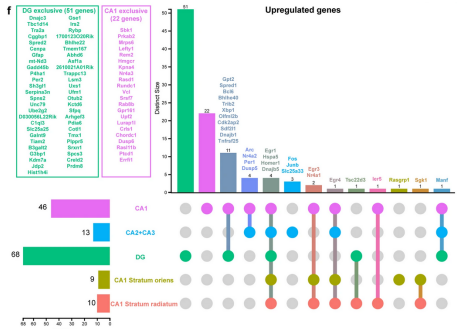


What genes and processes are involved in the different spatial subregions due to SOR training?

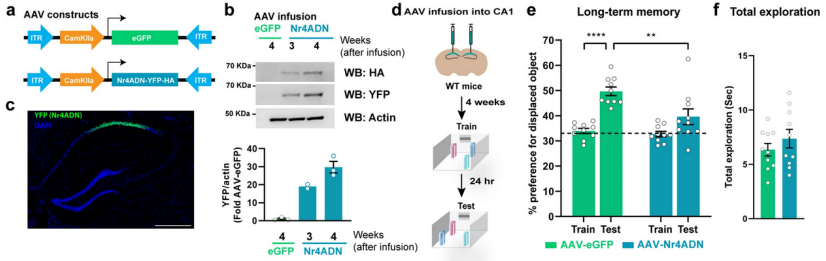


Functional relevance of subregion-specific expression of genes encoding the Nr4a family members

"Our spatial transcriptomics data revealed upregulation of all the three members of the Nr4a subfamily (Nr4a1, Nr4a2 and Nr4a3) in the CA1 pyramidal layer following learning (Fig. 3)."



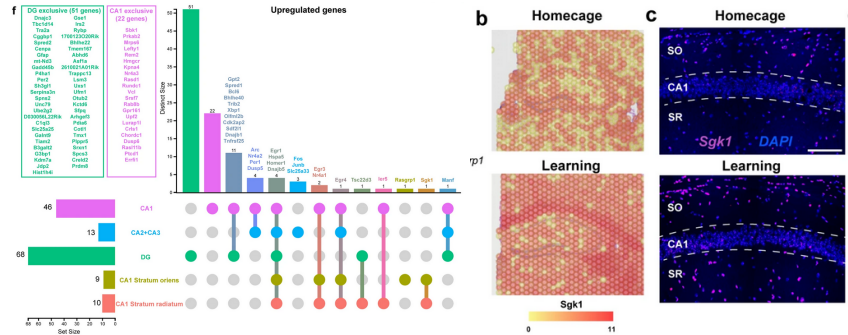
What happens to memory consolidation if transcription factor activity of Nr4a family members is inhibited in CA1?



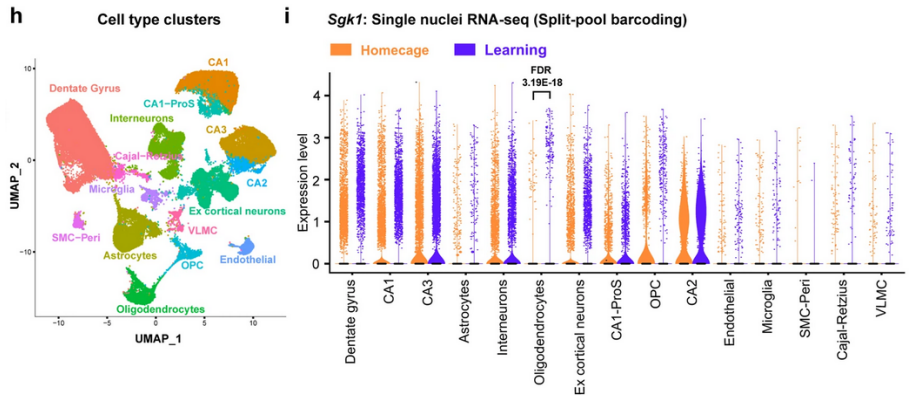
Inhibition in the dentate gyrus had no effect on memory consolidation (Supplementary Figure 8).

Sgk1 is induced in oligodendrocytes of CA1 stratum radiatum and oriens following learning

"Among the genes induced in stratum radiatum and oriens, protein kinase Sgk1 was the only upregulated gene appearing in both the stratum radiatum and oriens but not in the CA1 pyramidal layer (Fig. 5a, b)."



What cells types in the dorsal hippocampus are responsible for upregulation of Sgk1 after learning?



Homework